A Complexity-Theoretic Foundation for Metrology

Book I: Getting started with Type A Measurements

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Abstract

This textbook presents a novel theoretical framework connecting the physical process of measurement with the epistemological possibility of certifying its properties. It is designed for the theoretical metrologist, physicist, or engineer with a strong mathematical background but who may be new to the formalisms of computational complexity and information theory.

Through a curated cycle of research papers and pedagogical notes, we will embark on a journey to answer a fundamental question: **What properties must a measurement system possess to allow for a finite, deterministic proof about the infinite behavior of its output?**

Our central result is a proof of the necessary and sufficient conditions for such deterministic testability. We will show that for a measurement to be certifiable, the underlying process must be equivalent to a system possessing a **finite state** and a **continuous, finite-to-one model that causes a geometrically measurable information loss**. This work provides a rigorous, complexity-theoretic basis for understanding the structure of validatable measurements, particularly for Type A uncertainty analysis, and establishes the precise class of mathematical models and predicates that permit such validation.

Foreword and Guide to This Collection

The structure of this textbook is unconventional. It is not a monolithic narrative but a guided tour through a research project as it unfolded. We will present cleaned and annotated versions of seminal draft papers, interspersed with new tutorial chapters designed to bridge conceptual gaps. This approach has several advantages:

1. **Intellectual Honesty:** It shows the authentic progression of an idea—from initial intuitions and incomplete formulations to the subsequent refinements needed to make it rigorous.

2. **Pedagogy:** Foundational concepts are introduced precisely when they are needed to solve a problem that has just been posed.

3. **Modularity:** Each chapter and note is self-contained to a large degree, allowing the reader to focus on one piece of the puzzle at a time.

The reader is encouraged to follow the chapters in the order presented. We begin by establishing the core concepts and motivation (Part I), build to the central synthesis and its proofs (Part II), and conclude by consolidating the new framework and exploring its implications (Part III).

Welcome to the intersection of measurement, information, and computation.

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Part I
Foundational Concepts

This first part lays the groundwork for our theory. We begin by translating the core problems of computational complexity into the language of metrology to motivate the central question. We then develop the essential mathematical tools from information theory that will be required for the proof. The goal of this part is to equip the reader with the concepts and the "why" before diving into the formal proofs of the main thesis.

Chapter 1: An Introduction for the Metrologist

- The Foundational Gap: The difficulty of finding a property versus the ease of verifying it.
- A Lexicon of Computational Rigor: The classes **P** (Tractable), **NP** (Certifiable), and co-**NP** (Refutable).
- The Witness: Defining a "short proof" in the context of measurement science.
- The Intractability of "Forever": Why universal claims about infinite streams are generally beyond certification (Π_2^0-completeness).

Chapter 2: An Initial, Concrete Exploration

- **Title:** Tractability, Testability, and Tail Restrictions: A Formal Analysis
- A simple algebraic model: analyzing the effect of "clipping" rules on sample statistics.
- **Key Result:** For a *given* finite stream, verifying such statistical properties is in **P**.
- *Editor's Note:* This paper demonstrates how fixing the future makes a problem tractable and motivates the search for a truly short witness.

Chapter 3: The Language of Information: A Primer

- Probabilistic Information: Entropy, Mutual Information, and the link to invertibility.
- From Probabilistic to Geometric Loss: Introducing Metric Entropy.
- The Geometric Definition of Information Loss ($\Delta_\varepsilon^{\text{ent}}$) as the primary tool for proofs.

Part II
The Core Theory: Necessity and Sufficiency

Here we present the heart of the work. We construct a formal model of a measurement system and use the tools from Part I to prove our main results. We will demonstrate that the existence of a finite-state, information-losing model is both sufficient to enable certification and necessary for it.

Chapter 4: Formalizing the Measurement Process

- The Measurement Chain: Mapping the True Quantity (Q) to an Internal State (Y).
- The Stateful Generator: Evolving the Internal State into an Observable Stream ($Z_{1:\infty}$).
- The Role of Structural Randomness: Contrasting the model with i.i.d. processes.

Chapter 5: The Geometric Argument: Testability and its Limits

- The Model Regularity Assumption (Finite-to-One on a Compact Domain).
- The Sufficiency Proof: Showing that positive, bounded information loss enables NP-certification.
- The Hardness Proof: Showing that zero information loss leads to intractability.

Chapter 6: The Structure of Certifiable Measurement: The Guard-Band Equivalence

- **Central Concept:** The Guard-Band Predicate as the canonical form of any testable property.
- **Main Theorem:** A tail predicate admits a universal deterministic witness if and only if it is logically equivalent to a guard-band predicate.
- The necessity of a finite-state, information-losing model for certification.

Part III
Framework, Implications, and Future Work

In this final part, we step back to analyze the completed theory. We clarify its scope by unifying it with the familiar world of statistical uncertainty, and outline the new research questions it opens.

Chapter 7: From Statistical Confidence to Logical Certainty: A Unification of Frameworks

- One Object, Two Intents: Distinguishing between confidence ($\delta > 0$) and certainty ($\delta = 0$).
- The Black-Box vs. The Mechanistic View: How the demand for certainty forces a change in perspective.
- The Guard-Band as the Limit of a Confidence Interval.

Chapter 8: Conclusion: A New Foundation and Its Open Questions

- The Theory in a Nutshell: A recap of the logical chain from testability to model structure.
- Implications for Theoretical Metrology: A new language to define measurand and model validity.
- **Solved Problem:** The theoretical foundation for Type A analysis and the class of certifiable measurement models.
- **Open Questions:**
  - *Type B Uncertainty:* Extending the framework to handle non-statistical information.
  - *Constructive Foundations:* Building a practical software architecture based on this theory.
  - *Beyond Determinism:* Exploring interactive proofs for richer statistical claims.

Appendix A: Detailed Proof of Π_2^0-Hardness for Lossless Channels

A detailed, step-by-step reduction from the ‘TOTAL‘ problem to the lossless channel certification problem, formalizing the hardness argument from Chapter 5.

Chapter 1: An Introduction for the Metrologist

Dmytro Surdu

The purpose of measurement is not to produce numbers, but to produce statements of knowledge—statements that are rigorous, defensible, and hold predictive power. This chapter introduces a new lens through which to view the rigor of such statements: the lens of computational complexity theory.

1 The Foundational Gap: Finding vs. Verifying

In both metrology and computer science, there exists a profound and often frustrating gap between the difficulty of *finding* a result and the ease of *verifying* it once it has been found.

Example 1.1 (Deconvolution of a Sensor Response)**.** Consider a spectrometer whose output signal is a convolution of the true spectrum of a light source with the instrument's known, but complex, response function.

- **Finding the True Spectrum:** To find the true, underlying spectrum of the source, one must perform a deconvolution. This is a notoriously "ill-posed" problem. Small amounts of noise in the measured signal can lead to wildly oscillating, non-physical solutions for the true spectrum. Finding a stable, physically meaningful solution is a computationally intensive task, often requiring sophisticated regularization techniques and significant expertise. This is a hard "finding" problem.

- **Verifying a Proposed Spectrum:** Suppose a researcher provides a candidate spectrum and claims it is the true source. To verify this, one simply has to perform the forward operation: convolve the proposed spectrum with the instrument's response function. This is a straightforward, computationally stable, and fast calculation. If the result matches the measured signal within the noise tolerance, the claim has been *verified*.

This gap is the central subject of computational complexity theory. It formalizes this distinction through a set of "complexity classes," which are vast collections of problems that share a similar level of computational difficulty. For our purposes, three classes form the bedrock of our framework, acting as ideals of computational rigor.

2 A Lexicon of Computational Rigor

Let's abstract the idea of a "problem" to be a question that can be answered with a "yes" or "no". For example: "Does a spectrum exist that, when convolved, matches the measurement to within 0.1%?" We can categorize these problems based on how efficiently we can determine the answer.

2.1 P: The Class of the Tractable

Definition 2.1 (The Class **P**)**.** A decision problem is in the class **P** (for Polynomial Time) if there exists an algorithm that can solve it—find a "yes" or "no" answer—in a number of steps that is a polynomial function of the size of the input.

In our deconvolution example, the verification step—"Does this *given* candidate spectrum produce the correct output?"—is in **P**. The input is the candidate spectrum data, and the number of operations in a convolution is polynomial in that size.

Metrological Significance: Problems in **P** represent routine data processing. Calculating the mean, variance, or other statistics from a *finite, given* set of measurement data is a tractable, polynomial-time task. It is the realm of established, deterministic calculation.

2.2 NP: The Class of the Certifiable

This is the class of problems where, like deconvolution, a solution may be hard to find but is easy to check once you have a crucial piece of information, called a "witness."

Definition 2.2 (The Class **NP**)**.** A decision problem is in the class **NP** (for Nondeterministic Polynomial Time) if a "yes" instance of the problem has a short **witness** (or certificate) that can be verified in polynomial time.

The "shortness" requirement is key: the witness must be a compact piece of data, its length polynomially bounded by the size of the original problem description.

- **Problem:** "Does a physically meaningful spectrum exist that is consistent with the measured signal?"
- **Instance:** The measured signal data and the instrument response function.
- **Witness:** The proposed candidate spectrum itself. It is "short" as its data size is comparable to the measurement's.
- **Verification:** Perform the convolution and check the match. This is a **P**-time check.

Since a valid witness exists for any "yes" instance, this problem is in **NP**.

The Witness in Metrology: What is a witness in a metrological context? This is a central question of our work. A witness is a finite piece of data that allows a skeptic to become convinced of a property of a measurement system without having to trust the prover or repeat an entire, complex experiment. It is a compact, compelling piece of evidence.

For instance, to prove that a complex physical model can produce a signal that matches an observation, the witness might be the specific set of initial conditions that generates the desired signal.

Remark 2.1 (The P vs. NP Question)**.** A profound open question in all of science is whether $\mathbf{P} = \mathbf{NP}$. If they were equal, it would mean that any problem for which a solution can be easily verified can also be easily found. This seems unlikely. Most believe $\mathbf{P} \neq \mathbf{NP}$, meaning that problems like finding the deconvolution solution are genuinely harder than problems in $\mathbf{P}$.

2.3 co-NP: The Class of the Refutable

What if the answer to our question is "no"? How do we prove that *no physically meaningful spectrum* exists that can explain our data? The witness that worked for a "yes" answer (a single candidate spectrum) is useless. To prove "no," it seems you would have to show that every single possibility in an infinite space of functions fails. This is intractable.

This leads to the "co-class" of **NP**.

Definition 2.3 (The Class co-**NP**)**.** A decision problem is in co-**NP** if a "no" instance of the problem has a short witness that can be verified in polynomial time.

Proving a statement is false is equivalent to proving its negation is true. The problem "Does *no* spectrum exist..." is in co-**NP**. A "yes" answer to this negated problem is a "no" answer to the original.

Metrological Significance: Many claims in metrology are fundamentally about refutation. Consider the statement: "This dataset is consistent with our instrument model." A single, well-defined measurement that falls wildly outside the model's predictions can serve as a short, verifiable witness that the answer is "no." Thus, the problem of refuting the model is in **NP**. Conversely, the problem of *proving* the model's correctness—showing that no deviation exists or ever will exist—seems to require infinite evidence, placing it in the much harder co-**NP** realm.

3 The Intractability of "Forever": The Wall of Universal Quantification

The true difficulty in metrology arises when we make claims not about a finite dataset, but about the underlying data-generating process over an infinite future. Consider these two statements:

1. "The mean of the *first million samples* was within its specified tolerance."
2. "The running mean of the sample stream *will forever stay* within its specified tolerance."

Statement 1 is a **P** problem. We have the data; we do the calculation.

Statement 2 is of a different character entirely. It contains a **universal quantifier** over an infinite set: "**for all** future times $t > n$, property P holds." How could one possibly construct a finite witness for such a claim? You cannot provide the infinite tail of the data as a certificate.

This type of problem structure places it in a class high up in the "arithmetical hierarchy."

Definition 3.1 (The Class Π_2^0)**.** A set (or language) S is in Π_2^0 if its membership can be expressed in the form:

$$x \in S \iff \forall y \, \exists z \; R(x, y, z)$$

where R is a decidable predicate.

A statement like "for all future infinite tails 'v', the property 'P' holds" can often be cast into this form or a similar one (Π_1^0, which is co-r.e.). The details are technical, but the takeaway is profound: **These problems are provably outside of **NP** (and co-**NP**)**, unless the entire "Polynomial Hierarchy" of complexity classes collapses, which is considered extremely unlikely.

This means that for a generic, complex data-generating process, there is **no possibility** of a short, deterministic witness that can certify a property of its infinite tail.

4 The Central Question of This Work

We now arrive at the core motivation for this entire textbook. We have seen that statements about infinite behavior are, in general, beyond certification. Yet, in practice, metrologists and engineers make such claims of long-term stability and conformity all the time based on finite calibration data.

How is this possible?

Our answer will be that this is only possible when the measurement system itself is not a "generic, complex generator." It must have a special, simplifying structure. Our central thesis is that this special structure is a combination of a **finite internal state** and a **model that causes information loss**.

The following chapters will build a formal theory to show that these properties are the necessary and sufficient conditions that "collapse" the intractable Π_2^0 problem down into the realm of the certifiable (**NP**), and often even the tractable (**P**). It is what enables the existence of the very finite witnesses that rigorous measurement science implicitly relies on every day.

Chapter 2: An Initial, Concrete Exploration

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Editor's Note: *This chapter presents a verbatim reprint of an early research paper that initiated our inquiry. Its goal was to explore the boundary between tractable and intractable problems in a statistical setting. The paper focuses on a very concrete scenario: what can be said about the stability of a sample's mean and standard deviation when new data points are known to be "clipped," or confined to a band around the current mean?*

The main result (Theorem 6.2) is that if the entire *future data stream is given as input, the problem of checking its conformity is computationally trivial (in* $\mathbf{P}$*).*

*Pay close attention to the limitations discussed in the paper's conclusion and in our annotations. This work successfully simplifies the "universal quantifier" problem by assuming the future is fully known. This cleanly demonstrates the tractability of verification in a world without uncertainty, but it leaves the core metrological question unanswered: **How can we certify such a property when the future is unknown and we only have a finite witness?** This paper's limitations are the direct motivation for the more powerful tools we will develop in the subsequent chapters.*

Chapter 2: An Initial, Concrete Exploration

Tractability, Testability, and Tail Restrictions:
A Formal Analysis of Certificate–Based Verification Problems

Anonymous ChatGPT Collaboration

Original Draft Version

Abstract

We synthesise a sequence of results developed interactively in conversation, examining (i) the conceptual distinction between tractability and testability in computational complexity, (ii) the difficulty of certifying distributional hypotheses under tail–element restrictions, (iii) the position of such languages in the arithmetic and polynomial hierarchies, and (iv) a precise algebraic analysis of how small "clipped" additions to a sample affect its mean and unbiased standard deviation. We culminate by classifying the natural decision problem obtained from these statistics as lying in the class $\mathbf{P}$ (hence in $\mathbf{NP}$ but not $\mathbf{NP}$–complete unless $\mathbf{P} = \mathbf{NP}$).

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1 Introduction

The perennial question "Can we verify a claim more easily than we can find a proof of it?" permeates both theoretical computer science and applied statistics. In complexity theory this dichotomy manifests as the classes **P** (tractable, efficiently solvable) and **NP** (efficiently verifiable once a short witness is supplied). Statistics faces an analogous gap between determining whether data plausibly originate from a hypothesised distribution and confirming with confidence that a given sample obeys certain distributional properties.

In a series of informal conversations we investigated the interaction of these themes, guided by the following overarching questions:

1. What constitutes an **NP** witness for the statement "the datum x could have been generated by distribution q"?
2. When do tail restrictions on future samples suffice to collapse a universally quantified property into **NP** (or even **P**)?
3. How do elementary clipping rules affect fundamental sample statistics, and what complexity class contains the resulting decision problems?

This paper formalises and extends those results.

2 Preliminaries

2.1 Complexity Classes

We recall only those definitions required for later arguments; see [1] for a comprehensive treatment.

Definition 2.1 (Polynomial–time decidability)**.** A language L is in **P** if there exists a deterministic Turing machine M and a polynomial p such that for all x, M halts on input x in time $\leq p(|x|)$ and accepts iff $x \in L$.

Definition 2.2 (Verifiability – **NP**)**.** A language L is in **NP** if there exist a polynomial p and a deterministic verifier V such that $x \in L$ iff $\exists w \in \{0,1\}^{\leq p(|x|)}$ with $V(x,w) = \text{accept}$, and V runs in time $\leq p(|x|)$.

Definition 2.3 (Recursively Enumerable (r.e.))**.** A set $S \subseteq \Sigma^*$ is *recursively enumerable* if there exists a Turing machine that enumerates exactly the elements of S (possibly without halting). Equivalently, S is Σ_1^0 in the arithmetical hierarchy.

Throughout we adopt the usual inclusions $\mathbf{P} \subseteq \mathbf{NP} \subseteq \Sigma_1^0$.

2.2 Statistical Notation

For a finite sample $S_m = (s_1, \ldots, s_m)$ with $m \geq 2$ we write

$$\mu_m = \frac{1}{m}\sum_{k=1}^{m} s_k, \qquad \sigma_m^2 = \frac{1}{m-1}\sum_{k=1}^{m}(s_k - \mu_m)^2.$$

The unbiased sample standard deviation is $\sigma_m = \sqrt{\sigma_m^2}$.

3 Certificates for Distributional Support

Consider the language

$$L_{\text{support}} = \{(x, q) \mid q(x) > 0\}.$$

If q is specified by a probabilistic polynomial–time sampler $\texttt{Sample}_q(r)$, one obtains an **NP** witness simply by revealing the random tape r that satisfies $\texttt{Sample}_q(r) = x$. Verification reruns the sampler deterministically; see [2, Sec. 2].

4 Tail Restrictions and the Arithmetic Hierarchy

Let a fixed per–item predicate P define an admissible "clipping" band, e.g.

$$P(y) : |y - \mu_q| < 3\sigma_q.$$

Given a finite prefix $u \in \Sigma^*$, write

$$u \in L_P \iff \forall v \in \Sigma^\omega : P\big((u \cdot v)[i]\big) \text{ for all } i \geq |u|.$$

Because of the universal quantifier over an unbounded tail, L_P is a Π^0_2 set and therefore *not* recursively enumerable. Its complement — "there exists a violation of P" — is Σ^0_1 and hence r.e.

5 Effect of Clipping on Sample Statistics

We now turn to the algebraic core.

5.1 One–Step Update

Theorem 5.1 (Single Clipped Addition)**.** Let $a \geq 1$ and assume a new observation s_{n+1} obeys $|s_{n+1} - \mu_n| < a\sigma_n$. Then

$$|\mu_{n+1} - \mu_n| < a\sigma_n, \tag{1}$$

$$\sigma_{n+1} < a\sigma_n. \tag{2}$$

Proof. Equation (1) follows immediately from $\mu_{n+1} = \mu_n + \dfrac{s_{n+1} - \mu_n}{n+1}$. For the variance we have

$$\sigma^2_{n+1} = \left(1 - \frac{1}{n}\right)\sigma^2_n + \frac{n}{(n+1)^2}(s_{n+1} - \mu_n)^2,$$

which, under $|s_{n+1} - \mu_n| < a\sigma_n$, implies $\sigma^2_{n+1} < \sigma^2_n\big(1 - \frac{1}{n} + a^2\frac{n}{(n+1)^2}\big) \leq a^2\sigma^2_n$ whenever $a \geq 1$. □

5.2 Two–Step and Inductive Extension

Theorem 5.2 (Stability Under Repeated Clipping)**.** Let $a \geq 1$ and suppose s_{n+1}, s_{n+2} each satisfy the clip rule $|s_k - \mu_n| < a\sigma_n$. Then the enlarged sample S_{n+2} obeys (1) and (2) with $(n+1)$ replaced by $(n+2)$. Consequently, by induction, *every* finite extension assembled via rule P preserves the bounds.

Proof. Set $\delta_1 = s_{n+1} - \mu_n$ and $\delta_2 = s_{n+2} - \mu_n$. The new mean satisfies $|\mu_{n+2} - \mu_n| \leq \frac{|\delta_1|+|\delta_2|}{n+2} < \frac{2a\sigma_n}{n+2} < a\sigma_n$ for $n \geq 2$.

For the variance, writing $\mathrm{SSE}_m = (m-1)\sigma_m^2$ gives

$$\sigma_{n+2}^2 \leq \frac{(n-1)\sigma_n^2 + \delta_1^2 + \delta_2^2}{n+1} < \frac{n-1+2a^2}{n+1}\sigma_n^2 \leq a^2\sigma_n^2 \quad (a \geq 1).$$

□

6 Complexity of the Statistical Decision Problem

Definition 6.1 (Clipped–Sample Verification Language)**.** Fix $a > 1$ and $n \geq 2$. Let an instance be (S_n, S_i) with $i \geq n$. Define

$$L_{\text{clip}} = \big\{(S_n, S_i) \mid |\mu_i - \mu_n| < a\sigma_n \text{ and } \sigma_i < a\sigma_n\big\}.$$

Theorem 6.1. $L_{\text{clip}} \in \mathbf{P}$. Hence $L_{\text{clip}} \in \mathbf{NP}$ but is not $\mathbf{NP}$–complete unless $\mathbf{P} = \mathbf{NP}$.

Proof. **1. Input Model**

Let

$$S_i = (s_1, \ldots, s_i)$$

be an input instance, where each measurement

$$s_k = \frac{p_k}{q_k}$$

is a rational in lowest terms. Assume each numerator p_k and denominator q_k has bit-length at most B. Then the total input size is

$$N = i \times O(B).$$

2. Running-Sum Representation

We maintain two pairs of integers:

$$A_n = \sum_{k=1}^{n} p_k \Big(\prod_{j \neq k} q_j\Big), \qquad D_n = \prod_{k=1}^{n} q_k,$$

$$S_n = \sum_{k=1}^{n} \big(p_k \prod_{j \neq k} q_j\big)^2, \qquad D_n^2 = \big(D_n\big)^2.$$

Then

$$\sum_{k=1}^{n} s_k = \frac{A_n}{D_n}, \quad \sum_{k=1}^{n} s_k^2 = \frac{S_n}{D_n^2}.$$

We update these in each step:

$$\begin{aligned} A_{n+1} &= A_n \cdot q_{n+1} + p_{n+1} \cdot D_n, \\ D_{n+1} &= D_n \cdot q_{n+1}, \end{aligned} \qquad \begin{aligned} S_{n+1} &= S_n \cdot q_{n+1}^2 + \big(p_{n+1} \prod_{j=1}^{n} q_j\big)^2, \\ D_{n+1}^2 &= \big(D_{n+1}\big)^2. \end{aligned}$$

3. Bit-Length Growth and Operation Cost

After n steps:

$$\log_2 D_n = O(nB), \quad \text{and} \quad \text{bit-length}(A_n),\ \text{bit-length}(S_n) = O(nB).$$

Multiplication or addition of two L-bit integers on a Turing machine costs $O(L^2)$ time. Thus each update

$$(A_n, D_n) \to (A_{n+1}, D_{n+1}) \quad \text{and} \quad (S_n, D_n^2) \to (S_{n+1}, D_{n+1}^2)$$

takes

$$O\big((nB)^2\big)$$

time. Summing over $n = 1$ to i gives

$$\sum_{n=1}^{i} O\big((nB)^2\big) = O\big(i^3 B^2\big).$$

Since $N = iB$, this is $O(N^3)$.

4. Computing Mean and Variance

We obtain exact rationals:

$$\mu_i = \frac{A_i}{i\, D_i}, \quad \sigma_i^2 = \frac{i\, S_i - A_i^2/D_i}{i(i-1)\, D_i^2}.$$

Each involves a fixed number of multiplications, additions, and comparisons on $O(iB)$-bit integers, costing $O((iB)^2) = O(N^2)$ time.

5. Verifying the Clipped-Sample Inequalities

To decide membership in

$$L_{\text{clip}} = \{(S_n, S_i) \mid |\mu_i - \mu_n| < a\,\sigma_n,\ \sigma_i < a\,\sigma_n\},$$

we compute μ_n, σ_n and μ_i, σ_i as above, then perform two integer comparisons. Total time remains polynomial, specifically

$$O(N^3) + O(N^2) = O(N^3).$$

6. Conclusion

All steps—from updating running sums to testing the two inequalities—run in time polynomial in the input length N. Hence

$$L_{\text{clip}} \in \mathbf{P}.$$

□

7 Discussion and Future Work

The investigation highlights a recurring moral: *universal quantification over unseen data lifts decision problems out of* **NP** *unless one supplies a finite artefact that fixes the data.* In our setting the per–element clipping rule P constrains but does not determine the tail, so the language of "all tails good" lies in Π_2^0. Conversely, once the tail is given explicitly (as in L_{clip}) the problem collapses to **P**.

An immediate extension is to study *interactive* or *zero–knowledge* variants: can a prover convince a verifier that an *infinite* stream will forever obey P while revealing only $O(1)$ information? Such protocols inhabit the classes **AM**, **IP**, or **SZK** and merit further exploration.

Acknowledgements

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References

[1] M. Sipser. *Introduction to the Theory of Computation*, 3rd ed. Cengage, 2012.

[2] O. Goldreich. *Foundations of Cryptography, Vol. 2.* Cambridge University Press, 2008.

Chapter 3: The Language of Information: A Primer

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In the previous chapters, we established that certifying claims about an infinite future requires a "short witness." To construct such a witness, we must first understand how to quantify the knowledge it contains. This chapter introduces the mathematical language designed for this purpose: information theory. We will first explore the intuitive, probabilistic concepts of entropy and mutual information to establish the fundamental link between ***information loss*** *and functional* ***non-invertibility****. We will then introduce a more rigorous, geometric measure of information loss—metric entropy—which will serve as the primary tool for the proofs in the chapters to come.*

1 Probabilistic Information: Entropy and Mutual Information

At its core, information is the reduction of uncertainty. Before we can measure information, we must first have a way to measure uncertainty. In Shannon's information theory, the uncertainty of a random variable is called its **entropy**.

Remark 1.1 (Scope of this Section)**.** For the scope of our discussion on Shannon's mutual information, we assume that any continuous random variables have been quantized to a finite alphabet, so that the following definitions for discrete variables apply directly.

Definition 1.1 (Entropy)**.** The entropy $H(X)$ of a discrete random variable X that takes values $\{x_1, \ldots, x_n\}$ with probabilities $p_i = P(X = x_i)$ is defined as:

$$H(X) = -\sum_{i=1}^{n} p_i \log_2 p_i$$

The unit of entropy is the "bit." Entropy measures the average number of bits required to optimally encode a message drawn from the distribution of X.

If we observe another random variable Y, the remaining uncertainty about X is the **conditional entropy**, $H(X|Y)$. A key property is that knowledge can only reduce uncertainty on average: $H(X|Y) \leq H(X)$. This allows us to define the information that Y provides about X.

Definition 1.2 (Mutual Information)**.** The mutual information $I(X;Y)$ between two random variables X and Y is the reduction in the uncertainty of X due to knowing Y:

$$I(X;Y) = H(X) - H(X|Y)$$

1.1 Information Loss and Invertibility: The Crucial Link

In a measurement system, we are interested in what happens when an ideal reading X is processed by a deterministic function g to produce an output $Y = g(X)$. The Data Processing Inequality tells us that $I(Q;Y) \leq I(Q;X)$, meaning information about the original quantity Q can only be lost or preserved, never created.

The most important case for our theory is when information is perfectly preserved. When does $I(Q;X) = I(Q;Y)$? Or, more simply, when does the output Y contain all the information that was in the input X? This is equivalent to asking when $I(X;Y) = H(X)$. The answer establishes the fundamental connection between information preservation and functional invertibility.

Theorem 1.1 (MI Preservation $\iff$ Injectivity)**.** *Let X be a discrete random variable taking values in a finite set $\mathcal{X}$ with distribution P_X. Let $g : \mathcal{X} \to \mathcal{Y}$ be any deterministic function, and define the random variable $Y = g(X)$. Then the following are equivalent:*

$$I(X;Y) = H(X) \iff g \text{ is injective on the support of } P_X.$$

Injectivity on the support means that for every y with $P_Y(y) > 0$, there is exactly one $x \in \mathcal{X}$ such that $g(x) = y$ and $P_X(x) > 0$.

Proof. **1. Basic Identities.** Since Y is a deterministic function of X, the conditional entropy $H(Y|X) = 0$. From the chain rule for mutual information, $I(X;Y) = H(Y) - H(Y|X)$, we immediately have:

$$I(X;Y) = H(Y)$$

The condition $I(X;Y) = H(X)$ is therefore equivalent to $H(Y) = H(X)$.

From the chain rule for joint entropy, $H(X,Y) = H(Y) + H(X|Y)$ and also $H(X,Y) = H(X) + H(Y|X) = H(X)$. Equating these gives:

$$H(X|Y) = H(X) - H(Y)$$

Thus, the condition $H(X) = H(Y)$ is equivalent to $H(X|Y) = 0$. We have now shown that $I(X;Y) = H(X) \iff H(X|Y) = 0$. We will prove the theorem by proving $H(X|Y) = 0 \iff g$ is injective on the support.

2. ($\Rightarrow$): If $H(X|Y) = 0$, then g is injective on the support. The conditional entropy $H(X|Y)$ is a non-negative weighted average of the entropies of the conditional distributions $P_{X|Y=y}$:

$$H(X|Y) = \sum_{y \in \mathcal{Y}} P_Y(y) H(P_{X|Y=y})$$

Since each term is non-negative, the only way for the sum to be zero is if $H(P_{X|Y=y}) = 0$ for every y in the support of Y. The entropy of a discrete distribution is zero if and only if it is

a point mass (i.e., one outcome has probability 1 and all others have probability 0). Thus, for each such y, there must exist exactly one x with $P(X = x|Y = y) = 1$. This is precisely the definition of g being injective on the support of P_X.

3. ($\Leftarrow$): If g is injective on the support, then $H(X|Y) = 0$. Conversely, assume g is injective on the support of P_X. Then for any y in the support of Y, there is a unique x_y such that $g(x_y) = y$ and $P_X(x_y) > 0$. This means that if we observe $Y = y$, we know with certainty that X must have been x_y. The conditional distribution $P_{X|Y=y}$ is a point mass at x_y. The entropy of any point mass distribution is zero, so $H(P_{X|Y=y}) = 0$. Since this holds for all y in the support, the average $H(X|Y) = \sum P_Y(y) \cdot 0 = 0$.

This completes the proof. □

This theorem gives us our core intuition: preserving information is equivalent to preserving distinguishability. Losing information is equivalent to making distinct states indistinguishable—that is, non-invertibility.

2 Geometric Information Loss: Metric Entropy

Because Shannon-MI requires a distribution and a discrete (or quantized) space, we now recast the idea of information loss in purely geometric terms that are more suitable for our proofs involving continuous spaces. This tool is **metric entropy**.

Let our measurement map be the composite function $g \circ f : \mathcal{Q} \to \mathcal{Y}$. For any observed output $y \in \mathcal{Y}$, the set of all possible true states $q \in \mathcal{Q}$ that could have produced it is the **fibre** of y.

Definition 2.1 (Fibre)**.** The fibre of an observation $y \in \mathcal{Y}$ is the pre-image set:

$$\mathcal{G}_y = (g \circ f)^{-1}(y) = \{q \in \mathcal{Q} \mid g(f(q)) = y\}$$

If the map $g \circ f$ is invertible, every fibre is a single point. If it is non-invertible, some fibres will contain multiple points. The "information loss" is directly related to the "size" or "complexity" of these fibres. We measure this using covering numbers.

Definition 2.2 (ε-Covering Number / Metric Entropy)**.** Let (S, d) be a subset of a metric space. The **ε-covering number** of S, denoted $N_\varepsilon(S)$, is the minimum number of closed balls of radius ε needed to completely cover S. The quantity $\log_2 N_\varepsilon(S)$ is often called the metric entropy of S.

We can now provide our primary, purely geometric definition of information loss.

Definition 2.3 (Geometric Information Loss)**.** The **geometric information loss** of the channel $g \circ f$ at scale ε is the worst-case metric entropy over all possible fibres:

$$\Delta_\varepsilon^{\text{ent}} = \sup_{y \in \mathcal{Y}} \log_2 N_\varepsilon(\mathcal{G}_y)$$

2.1 The Power of the Geometric Definition

This new definition is superior for our theoretical purposes for several key reasons:

1. **It is Analyst-Independent:** It depends only on the function $g \circ f$ and the metric d on the space $\mathcal{Q}$. It does not depend on any probability measure P_Q or any artificial quantization of the output space.

2. **It is a Worst-Case Measure:** By using the supremum, $\Delta_\varepsilon^{\text{ent}}$ directly gives us a bound that holds for *every* fibre, which is exactly what we need for the worst-case guarantees of complexity theory.

3. **It Connects Perfectly to Invertibility:**

   - If $g \circ f$ is **invertible**, every fibre $\mathcal{G}_y$ is a singleton. Thus $N_\varepsilon(\mathcal{G}_y) = 1$ for all y, and $\Delta_\varepsilon^{\text{ent}} = \log_2(1) = 0$. This is the **information-lossless** case.
   - If $g \circ f$ is **non-invertible**, at least one fibre contains multiple points. If these points have a minimum separation, then for small enough ε, $N_\varepsilon(\mathcal{G}_y) \geq 2$ for some y, which implies $\Delta_\varepsilon^{\text{ent}} \geq 1$. This is the **information-losing** case.

In the chapters that follow, we will exclusively use this geometric definition, $\Delta_\varepsilon^{\text{ent}}$, to prove our main theorems. It provides the clean, rigorous, and powerful tool we need to build our theory of certifiable measurement.

Chapter 4: Formalizing the Measurement Process

Dmytro Surdu

Having established our motivation and the necessary information-theoretic tools, we now construct the central object of our analysis: a formal, mathematical model of a measurement process. The goal of this chapter is to create a model that is simple enough for rigorous analysis, yet rich enough to capture the essential features of a real-world measurement system—particularly the flow of information from an unknown physical quantity to a final, observable data stream. This "Measurement Chain" will serve as the canvas for all the theorems in Part II.

1 The Deterministic Model: Front-Loading Randomness

To analyze the logical structure of certification, we must first isolate the sources of uncertainty. We model a measurement system as a process whose behavior is fully determined by its initial conditions. All of the randomness or uncertainty in this model is "front-loaded" into the one-time choice of the true physical state of the object being measured. Once that state is fixed, the entire future evolution of the measurement's output is also fixed.

This is a powerful and necessary simplification. It allows us to distinguish between two types of systems:

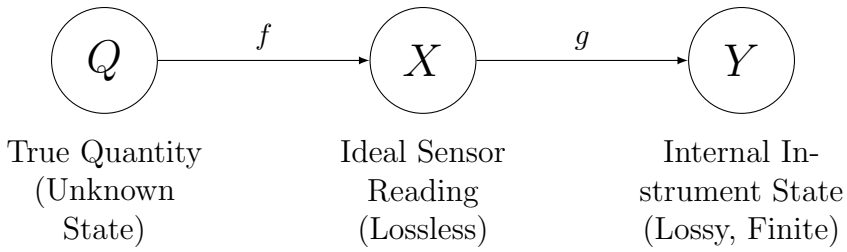
1. **Systems with Ongoing Randomness:** Where each new data point involves a fresh, independent random event (e.g., thermal noise). As we saw in Chapter 1, certifying the infinite behavior of such systems is generally intractable.

2. **Systems with Structural Randomness:** Where the entire infinite output stream is a deterministic function of a single, initial random state.

Our theory is a theory of the second type of system. It seeks to answer the question: *What structural properties of the instrument itself make certification possible in a world where all randomness is contained at the origin?*

2 The Measurement Chain: From Quantity to State

We model the initial act of measurement as a chain of transformations that maps the true physical quantity to a finite, internal state of the measurement instrument.



Let's define each component of this initial chain formally.

2.1 Q: The True Quantity (The Unknown State)

Definition 2.1. The **True Quantity**, Q, is a random variable representing the complete physical state of the object being measured. It takes a specific value q from a separable metric space $(\mathcal{Q}, d)$. We assume q is drawn from some unknown but fixed probability distribution P_Q.

The state q is the "thing in itself." It encapsulates every property of the object—its primary value, its temperature, its impurities, its history—that could possibly influence the measurement outcome.

2.2 f: The Ideal Sef: The Ideal Sensor Mappingnsor Mapping

Definition 2.2. The **Ideal Sensor Mapping**, $f : \mathcal{Q} \to \mathcal{X}$, models the lossless physics of the sensor. It maps the true physical state q to an "ideal" representation $x = f(q)$ in some internal space $\mathcal{X}$ (e.g., a perfect, infinite-precision voltage).

We assume f is **injective** (one-to-one), meaning that if $q_1 \neq q_2$, then $f(q_1) \neq f(q_2)$. This ensures that at this stage, no information about Q has been lost. The ideal reading $X = f(Q)$ is a perfect, lossless proxy for the true state Q.

2.3 g: The Lossy Model (The Information-Losing Stage)

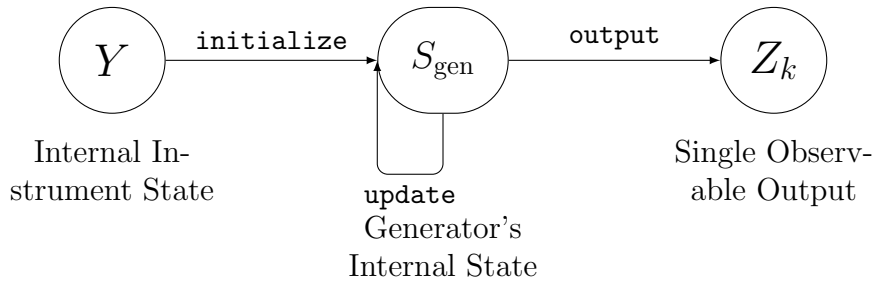
Definition 2.3. The **Lossy Model**, $g : \mathcal{X} \to \mathcal{Y}$, represents the first stage where information about Q is irreversibly lost. It maps the ideal reading x to a final, coarsened internal state $y = g(x)$.

This function is the most important part of the measurement model. It represents physical or computational processes like quantization, clipping, saturation, or feature extraction. Its key property is that it is **non-injective**. Multiple different ideal readings, x_1 and x_2, can be mapped to the same internal state, $y = g(x_1) = g(x_2)$. As established in Chapter 3, this non-invertibility is precisely what causes a quantifiable loss of information about the original state Q.

The output of this chain, $Y = g(f(Q))$, is the final, finite state of the instrument after the measurement act is complete. It contains all the information about Q that the instrument has successfully captured.

3 The Stateful Generator: From State to Stream

Once the instrument has captured the state Y, it generates the observable data stream. We model this as a separate, deterministic, stateful process—a transducer or Mealy machine.



Definition 3.1 (The Stateful Generator)**.** The stream generator is a deterministic transducer defined by:

1. A set of internal generator states, $\mathcal{S}_{\text{gen}}$.
2. An initialization function, initialize : $\mathcal{Y} \to \mathcal{S}_{\text{gen}}$, which sets the initial generator state $s_0 = \text{initialize}(y)$.
3. A transition function, $H : \mathcal{S}_{\text{gen}} \to \mathcal{S}_{\text{gen}} \times \mathcal{Z}$, that takes the current generator state s_k and produces the next state s_{k+1} and the next observable output z_{k+1}.

The full observable stream is the sequence of outputs $Z_{1:\infty} = (z_1, z_2, z_3, \dots)$ produced by the iterative application of H, starting from s_0.

This more rigorous model is crucial. It separates the initial, information-losing "measurement" from the subsequent, deterministic "processing." It also provides the formal machinery needed to discuss concepts like a system "attractor," which is a property of the dynamics of the state s_k under repeated application of the transition function H.

3.1 The Complete Model

Combining these stages, our complete model for a certifiable measurement system is a composite process: 1. **Measurement Act:** The true state q is mapped to a finite internal instrument state y via the potentially lossy function $g \circ f$. 2. **Stream Generation:** The state y initializes a deterministic stateful transducer, which then autonomously generates the infinite observable data stream $Z_{1:\infty}$.

The entire stream is thus a deterministic function of the initial state q, which we can write as $Z_{1:\infty} = \text{Generate}(\text{initialize}(g(f(q))))$.

This model is the theoretical arena for our work. It is within this structure that we can formally prove our central theorems about the relationship between information loss, state, and testability. We will now proceed to show how the non-injectivity of the function g creates geometric structures in the space $\mathcal{Q}$ that are the key to constructing the short, powerful witnesses we seek.

Chapter 5: The Geometric Argument: Testability and its Limits

Dmytro Surdu

Editor's Note: *This chapter presents the first major synthesis of our theory. Using the Measurement Chain model from Chapter 4 and the geometric tools from Chapter 3, this paper builds the complete information-theoretic case for our thesis. This is a revised and polished version of the original draft, now employing the more powerful metric-entropy-based arguments and a more precise definition of an admissible predicate.*

It accomplishes three critical tasks:

1. ***It formalizes the witness.*** *The certificate for a property is defined as a '(coarse-label, fibre-index)' pair, which serves to pin down the hidden initial state of the system.*

2. ***It proves the positive result (sufficiency).*** *The main theorem demonstrates that if a measurement model has positive geometric information loss ($\Delta_\varepsilon^{ent} > 0$) and satisfies a reasonable regularity condition, then tail properties can be certified by a short, polynomial-size* **NP** *witness.*

3. ***It proves the negative result (necessity of loss).*** *It shows that if the model is perfectly lossless ($\Delta_\varepsilon^{ent} = 0$), the certification problem becomes provably intractable, placing it outside of* **NP**.

This chapter provides the core "if and only if" argument, linking the geometric property of information loss directly to the logical property of testability.

Abstract

We formalise and prove an "information–loss ⇔ testability" theorem for deterministic measurement chains. We define information loss in purely geometric terms using the metric entropy of the system's state-space fibres. We show that a positive, uniformly bounded information loss—a property of non-invertible and regular (continuous, finite-to-one) models—is a sufficient condition for tail properties of the measurement stream to be certifiable in the complexity class **NP**. Conversely, we show that a lossless channel leads to a provably intractable (Π_2^0-hard) certification problem. The argument is constructive, yielding an explicit finite witness that consists of a coarse system state and an index within the information-loss fibre.

1 Introduction

Complexity theory distinguishes problems that are merely *testable* (**NP**) from those that seem to require intractable proof (co-**NP** and harder). In metrology, we face an analogous gap: certain statements about a system can be certified from finite data, while others seem to require impossible knowledge of an infinite future. This paper proves that the bridge between these two worlds is **information loss**.

We will demonstrate that a deterministic measurement system permits a finite, polynomial-size certificate for a tail property if and only if the measurement channel is non-invertible. We formalize this non-invertibility as a positive, uniformly bounded geometric information loss, and show that the certificate is precisely a state-space coordinate: a coarse label plus an index locating the true micro-state within its collapsed fibre.

Our argument proceeds as follows. In Section 2, we define the necessary regularity conditions for a model and a predicate. In Section 3, we define the state witness, prove our main sufficiency theorem (Theorem 3.1), and then prove the corresponding hardness theorem for the lossless case (Theorem 3.2).

2 Framework

We adopt the Measurement Chain model defined in Chapter 4, with its components Q, f, g, and the resulting stream $Z_{1:\infty}$. We also use the definitions of the fibre $\mathcal{G}_y = (g \circ f)^{-1}(y)$ and geometric information loss $\Delta_\varepsilon^{\text{ent}} = \sup_{y \in \mathcal{Y}} \log_2 N_\varepsilon(\mathcal{G}_y)$ from Chapter 3.

2.1 Model Regularity

To ensure our results hold in the worst-case, we require a regularity condition on the measurement map $g \circ f$. This condition ensures the instrument's uncertainty structure is well-behaved and non-pathological.

Definition 2.1 (Model Regularity (Finite-to-One on a Compact Domain))**.** The measurement map $g \circ f$ satisfies **Model Regularity** if:

1. The space of true quantities, $\mathcal{Q}$, is compact under the metric d.
2. The map $g \circ f : \mathcal{Q} \to \mathcal{Y}$ is Lipschitz on $\mathcal{Q}$.
3. The map is **finite-to-one**: there exists an integer M such that for every $y \in \mathcal{Y}$, the fibre $\mathcal{G}_y$ has cardinality $|\mathcal{G}_y| \leq M$.

Lemma 2.1 (Consequence of Model Regularity)**.** *If a model satisfies the regularity condition, then its geometric information loss Δ_ε^{ent} is uniformly bounded. For all sufficiently small ε, we have $N_\varepsilon(\mathcal{G}_y) = |\mathcal{G}_y| \leq M$ for every fibre y. Therefore,*

$$\Delta_\varepsilon^{ent} \leq \log_2 M$$

The information loss is bounded by a constant determined by the model's structure.

2.2 Admissible Predicates

Our results apply to tail predicates that are "well-behaved" enough to be decided by a finite simulation and robust to small errors. This captures the requirement that the predicate must eliminate all residual randomness in finite time.

Definition 2.2 (Admissible Tail Predicate)**.** A predicate P_{tail} on infinite streams $Z_{1:\infty}$ is **admissible** if there exist a polynomial $T(n)$ and a deterministic Boolean function Φ such that for any true state $q \in \mathcal{Q}$ and any ε-approximation q^* with $d(q, q^*) \leq \varepsilon$:

1. **Finite-Window Determination:** The truth of the predicate is determined by a finite portion of the tail stream:

$$P_{\text{tail}}(Z_{1:\infty}) = \Phi(Z_{n+1}, \ldots, Z_{n+T(n)})$$

2. **ε-Robustness:** Small errors in the initial state do not flip the decision. Let Z be the stream from q and Z^* be the stream from q^*. Then:

$$\Phi(Z_{n+1}, \ldots, Z_{n+T(n)}) = \Phi(Z^*_{n+1}, \ldots, Z^*_{n+T(n)})$$

3 The Main Results: Testability and its Limits

3.1 Sufficiency: Information Loss Enables NP-Certification

Definition 3.1 (State Witness)**.** Fix a precision ε. For any true state $q \in \mathcal{Q}$, its **state witness** is the pair $w(q) = (y, r)$, where:

- $y = g(f(q))$ is the coarse instrument state (the fibre label).
- r is the index of the ε-ball containing q within a pre-defined, canonical cover of the fibre $\mathcal{G}_y$.

Lemma 3.1 (Uniform Witness-Length Bound)**.** *Under Model Regularity (compact $\mathcal{Q}$, Lipschitz $g \circ f$, finite-to-one with bound M), for any precision $\varepsilon > 0$ the state witness*

$$w(q) \;=\; \big(y,\; r\big)$$

where $y = g(f(q))$ and r is the index of the ε-ball containing q in a fixed canonical cover of the fibre $\mathcal{G}_y$, has bit-length

$$|w(q)| \;\leq\; \big\lceil \log_2 N_{\mathcal{Y}}(\varepsilon) \big\rceil \;+\; \lceil \log_2 M \rceil \;=\; O\big(\log(1/\varepsilon)\big) \;+\; \log_2 M.$$

Here $N_{\mathcal{Y}}(\varepsilon)$ is the size of a minimal ε-cover of $\mathcal{Y} = g(f(\mathcal{Q}))$.

Proof. **1. Total boundedness of $\mathcal{Y}$.** Since $\mathcal{Q}$ is compact and $g \circ f$ is Lipschitz, its image $\mathcal{Y} = (g \circ f)(\mathcal{Q})$ is also compact and hence totally bounded. By definition of total

boundedness, for each $\varepsilon > 0$ there exists a finite set of "centers" $\{y_1, \ldots, y_{N_{\mathcal{Y}}(\varepsilon)}\} \subset \mathcal{Y}$ such that

$$\mathcal{Y} \subset \bigcup_{j=1}^{N_{\mathcal{Y}}(\varepsilon)} B\big(y_j, \varepsilon\big).$$

We fix once and for all this canonical ε-cover and agree on a public indexing $j = 1, \ldots, N_{\mathcal{Y}}(\varepsilon)$.

2. Encoding the coarse label y. Given $q \in \mathcal{Q}$, let $y = g(f(q))$. By the cover property, there is at least one index j with

$$d\big(y, y_j\big) \;\leq\; \varepsilon.$$

Choosing one (e.g. the smallest such j) identifies the ε-ball containing y. Encoding this index $j \in \{1, \ldots, N_{\mathcal{Y}}(\varepsilon)\}$ in binary requires

$$\lceil \log_2 N_{\mathcal{Y}}(\varepsilon) \rceil$$

bits. By standard estimates for metric-entropy, $N_{\mathcal{Y}}(\varepsilon) = O(\varepsilon^{-d})$ in many spaces, so $\log_2 N_{\mathcal{Y}}(\varepsilon) = O(\log(1/\varepsilon))$.

3. Encoding the fibre index r. Under the finite-to-one assumption, each fibre $\mathcal{G}_y$ has size at most M. We fix a public ordering of its elements and let $r \in \{1, \ldots, |\mathcal{G}_y|\}$ denote the position of the true state q. Encoding r takes at most

$$\lceil \log_2 M \rceil$$

bits, which is a constant independent of ε.

4. Total bit-length. Concatenating the two binary codes for j and r yields the full witness $w(q)$. Hence

$$|w(q)| \;\leq\; \lceil \log_2 N_{\mathcal{Y}}(\varepsilon) \rceil \;+\; \lceil \log_2 M \rceil \;=\; O\big(\log(1/\varepsilon)\big) \;+\; \log_2 M,$$

as claimed. □

Theorem 3.1 (Sufficiency)**.** *Under Model Regularity, fix a precision ε such that the geometric information loss $\Delta_\varepsilon^{ent} > 0$ (i.e., $M > 1$). Let P_{tail} be any admissible tail predicate. Then the language of "good heads" L_P is in the complexity class* **NP**.

Proof. Our NP-certificate is precisely the state witness $w(q) = (y, r)$ corresponding to the true initial state q. Its length is polynomially bounded by the previous lemma. The verifier, on input head $z_{1:n}$ and witness (y, r), performs the following polynomial-time algorithm:

1. Reconstruct the center $q^\star$ of the r-th ball in the public cover of fibre $\mathcal{G}_y$. This $q^\star$ is an ε-approximation of the true state q.

2. **Check Consistency:** Verify that simulating the stream generator starting from $q^\star$ for n steps reproduces exactly the observed head $z_{1:n}$. If not, reject.

3. **Verify Predicate:** Simulate the stream generator, starting from $q^\star$, for the next $T(n)$ steps (the horizon from the admissibility definition) to get the tail segment $z^\star_{n+1:n+T(n)}$.

4. Evaluate the Boolean function $\Phi(z^\star_{n+1:n+T(n)})$. Accept if it returns true, reject otherwise.

Because the predicate is ε-robust, the conclusion drawn from the simulation of $q^\star$ is the same as the one for the true state q. The verifier is deterministic, runs in polynomial time, and correctly accepts all true instances. Therefore, the language L_P is in **NP**. □

3.2 Necessity: Lossless Channels are Intractable

We now show that information loss is not just sufficient, but necessary. If the channel is lossless, the problem becomes provably intractable.

Theorem 3.2 (Hardness)**.** *Let the measurement map $g \circ f$ be a continuous bijection (i.e., it is invertible). This implies the channel is lossless, with $\Delta^{ent}_\varepsilon = 0$ for all $\varepsilon > 0$. For any non-trivial admissible tail predicate P_{tail}, the language of "good heads" L_P is Π^0_2-hard.*

Sketch. The full proof is detailed in the Appendix. The core idea is a reduction from the known Π^0_2-hard problem 'TOTAL'. We construct a specific stream generator h_e whose behavior depends on a Turing machine index e, and a predicate P_e that is true if and only if M_e halts on all inputs.

The key step is this: since $g \circ f$ is a bijection, the observed head $z_{1:n}$ uniquely determines the entire infinite stream's initial state q. There is no ambiguity for a witness to resolve. The verifier is therefore faced with the full, unresolvable task of verifying the universal quantification 'for all inputs x...' required by the 'TOTAL' reduction. This task is known to be Π^0_2-hard. Since we can reduce any instance of 'TOTAL' to an instance of our problem, our problem must be at least as hard. It therefore cannot be in **NP** (unless the Polynomial Hierarchy collapses). □

4 Conclusion

The geometric properties of a measurement map determine its certificability. A non-invertible map that satisfies Model Regularity induces a positive, bounded geometric information loss. This loss creates the structure necessary for a short, finite witness to exist, placing the certification problem in **NP**. Conversely, a perfectly lossless (invertible) map offers no such structure, leaving the certification problem in the intractable realm of the higher arithmetic hierarchy. Information loss is therefore the fundamental enabler of deterministic testability.

Chapter 6: The Structure of Certifiable Measurement: The Guard-Band Equivalence

Dmytro Surdu

Editor's Note: *In this chapter we prove that any tail property allowing a polynomial-size, deterministic witness must—and only can—be a guard-band predicate.*

1 Motivation: From a Sufficient Condition to a Universal Law

We have shown that non-invertible (information-losing) maps admit short witnesses. Is there any other mechanism to certify a tail property? We answer: no. Any deterministic, streaming verifier that makes an irrevocable accept/reject decision on each sample can only test a per-sample "guard-band" predicate, and vice versa. This chapter formalizes this claim into a necessary and sufficient condition, providing a complete characterization of deterministic testability.

2 The Guard-Band Predicate

A guard-band predicate is a property asserting that every sample in the infinite tail of a measurement stream must lie within a specific acceptance region. The crucial feature is that this region can depend on the underlying true state of the system, Q.

Definition 2.1 (Guard-Band Predicate)**.** A tail predicate P_{tail} is a **guard-band predicate** if it is logically equivalent to a statement of the form:

$$\forall k > n,\ Z_k \in \mathcal{R}_{\theta_Q}$$

where:

1. $\theta_Q = T(P_Q)$ is a finite-dimensional vector of statistics derived from the true (unknown) distribution P_Q. The mapping $T : \mathcal{P}(\mathcal{Q}) \to \Theta \subseteq \mathbb{R}^d$ is Lipschitz continuous.

2. The map $\theta \mapsto \mathcal{R}_\theta$, from the space of statistics Θ to acceptance regions, is Lipschitz continuous in the Hausdorff metric.

Intuitively, the system's true nature determines a parameter θ_Q, and this parameter in turn sets a fixed "lane" or "channel" $\mathcal{R}_{\theta_Q}$ in which all future measurements must remain.

Definition 2.2 (Guard-Band Region Regularity and Margin)**.** For a given parameter θ and its associated symbol-level acceptance region $\mathcal{R}_\theta \subseteq \mathcal{Z}$, define the *margin* of $\mathcal{R}_\theta$ as

$$\delta(\mathcal{R}_\theta) := \inf_{z \in \mathcal{R}_\theta} \operatorname{dist}(z, \mathcal{Z} \setminus \mathcal{R}_\theta),$$

where the distance is taken in the ambient metric on $\mathcal{Z}$ inherited from the measurement/state space. We say that $\mathcal{R}_\theta$ satisfies the *minimal regularity* condition if

$$\delta(\mathcal{R}_\theta) > 0,$$

i.e., $\mathcal{R}_\theta$ contains a nontrivial open buffer around each of its points and is not "knife-edge" thin. Equivalently, $\mathcal{R}_\theta$ is the union of balls of radius at least $\delta(\mathcal{R}_\theta)$ contained in itself.

This is the weakest possible regularity needed for finite-horizon truncation: it guarantees that small perturbations (up to size less than the margin) of an accepted symbol remain accepted, providing the slack required in contraction-based error accumulation.

3 The Main Equivalence Theorem

We now state the main equivalence and prove each direction, starting with establishing the operational definitions.

Remark 3.1 (Metric Notation)**.** In this chapter we write $d_\Theta(\theta, \theta')$ for the metric on Θ. Equivalently, if Θ carries a norm $\|\cdot\|$, then

$$d_\Theta(\theta, \theta') \;=\; \|\theta - \theta'\|.$$

Similarly, d_H denotes the Hausdorff distance on subsets of $\mathcal{Z}$.

3.1 State Transition and Emission Definition

Recall from Chapter 4 that our measurement model consists of:

- A compact "true-state" space $(\mathcal{Q}, d_Q) \subset \mathbb{R}^m$.
- An *ideal sensor* $f : \mathcal{Q} \to \mathcal{X}$ which is injective and Lipschitz.
- A *lossy model* $g : \mathcal{X} \to \mathcal{Y}$ which is finite-to-one and Lipschitz.
- A *stream generator* (Mealy transducer)

  $$H : \mathcal{S}_{\text{gen}} \;\longrightarrow\; \mathcal{S}_{\text{gen}} \times \mathcal{Z},$$

  initialized by $s_0 = \text{initialize}(y)$ with $y = g\big(f(q)\big)$.

We now define the induced *statistic-space* and its update/emission maps:

- $\Theta = T\big(\mathcal{P}(\mathcal{Q})\big) \subset \mathbb{R}^d$, the finite-dimensional parameter space, where

$$T : \mathcal{P}(\mathcal{Q}) \longrightarrow \Theta$$

  is assumed *Lipschitz continuous* (Guard-Band Def. 1).

- The coordinate projections $\pi_1, \pi_2 : \mathcal{S}_{\text{gen}} \times \mathcal{Z} \to \mathcal{S}_{\text{gen}}, \mathcal{Z}$ and the induced maps

$$U = \pi_1 \circ H : \mathcal{S}_{\text{gen}} \to \mathcal{S}_{\text{gen}}, \qquad E = \pi_2 \circ H : \mathcal{S}_{\text{gen}} \to \mathcal{Z}.$$

  We identify $\mathcal{M} = \mathcal{S}_{\text{gen}}$ with the belief-state space via the map $\theta \mapsto s$ (i.e. push T-statistics forward to generator states).

- By Model Regularity (Ch 4) and finite-to-one of g, $\mathcal{S}_{\text{gen}}$ is compact, and since H is continuous (indeed Lipschitz if f, g and initialize are), it follows that

$$U : \mathcal{M} \to \mathcal{M} \quad \text{and} \quad E : \mathcal{M} \to \mathcal{Z}$$

  are both *Lipschitz continuous.*

- Finally, from the Guard-Band Predicate (Def. 6.1) the map $\theta \mapsto \mathcal{R}_\theta$ is Lipschitz in the Hausdorff metric on subsets of $\mathcal{Z}$.

Summary of Derived Assumptions.

(A1) $\mathcal{M} = \mathcal{S}_{\text{gen}}$ is a compact metric space.

(A2) $U : \mathcal{M} \to \mathcal{M}$ is Lipschitz continuous.

(A3) $E : \mathcal{M} \to \mathcal{Z}$ is Lipschitz continuous.

(A4) $\Theta \subset \mathbb{R}^d$ is compact and $T : \mathcal{P}(\mathcal{Q}) \to \Theta$ is Lipschitz.

(A5) $\theta \mapsto \mathcal{R}_\theta$ is Lipschitz in the Hausdorff metric.

In addition, we impose a local contraction assumption on U:

(A6) **Local Contraction:** There exists a *true* parameter $\theta_Q \in \Theta$, a radius $r > 0$, and $\rho \in [0, 1)$ such that for all $z \in \mathcal{Z}$ and all $\theta, \theta' \in B_r(\theta_Q)$,

$$d_\Theta\big(U(\theta, z),\, U(\theta', z)\big) \;\le\; \rho\, d_\Theta(\theta, \theta').$$

Clarification. The term "true value" $\theta_Q := T(P_Q)$ is used as a canonical representative of the underlying process—it is not to be interpreted as an epistemic absolute that the verifier uniquely recovers. In the equivalence, a certificate may select any statistic θ (possibly distinct from θ_Q) whose induced guard-band behavior is equivalent (within the permitted stability/Lipschitz slack) to that of the canonical choice. Lipschitz continuity and the local contraction/stability assumptions ensure that such witness-chosen θ and the canonical θ_Q produce guard-band predicates that differ only by controlled, negligible perturbations; thus, the equivalence holds with respect to any such representative without requiring access to an inaccessible "true" metaphysical value.

3.2 The True-Value / Sufficient-Statistic Axiom

In Chapter 2, we saw a concrete "mean and sd" example:

$$\theta_n = (\mu_n, \sigma_n), \quad \theta_{n+1} = U(\theta_n, s_{n+1}) = \begin{pmatrix} \dfrac{n}{n+1}\,\mu_n + \dfrac{1}{n+1} s_{n+1}, \\ \sqrt{\left(1 - \frac{1}{n}\right)\sigma_n^2 + \frac{n}{(n+1)^2}(s_{n+1} - \mu_n)^2} \end{pmatrix}.$$

A direct calculation showed

$$\|\theta_{n+1} - \theta'_{n+1}\| \;\le\; \frac{n}{n+1}\,\|\theta_n - \theta'_n\|,$$

so the update was a strict contraction $\rho = \frac{n}{n+1} < 1$. That bound was the keystone of our finite-horizon proof.

Generalization candidates. One might ask: can we replace "mean and sd" by any other finite-dimensional statistic T? The answer is yes, *provided* its one-step recursion

$$\theta_{n+1} = U\big(\theta_n,\ z_{n+1}\big)$$

satisfies a *uniform contraction* in a neighborhood of the true parameter. Examples include:

- *Exponentially weighted moving averages:* $\theta_{n+1} = \alpha\,\theta_n + (1-\alpha) z_{n+1},\ \alpha < 1$.
- *Polynomial-regression coefficients:* update via recursive least-squares with forgetting factor < 1.
- *Kernel-density modes:* approximate Newton-Raphson steps with step-size < 1.

In each case a simple Jacobian or spectral-radius check shows there is an $r > 0$ and $\rho < 1$ so that

$$d\big(U(\theta, z),\, U(\theta', z)\big) \;\le\; \rho\, d(\theta, \theta'), \quad \forall\, \theta, \theta' \in B_r(\theta_Q),\ z \in \mathcal{Z}.$$

Necessity of local contraction. Without such a contraction bound, small errors in the witness-statistic can amplify under iteration and breach the guard-band, invalidating any finite-horizon check.

Axiom 3.1 ("True-Value"/Sufficient-Statistic Axiom)**.** *There exists a (unknown) "true" parameter $\theta_Q \in \Theta$, a radius $r > 0$, and a constant $\rho \in [0, 1)$ such that for every input $z \in \mathcal{Z}$ and all $\theta, \theta' \in B_r(\theta_Q)$:*

$$d\big(U(\theta, z),\, U(\theta', z)\big) \;\le\; \rho\, d(\theta, \theta').$$

In other words, once the statistic lies within r of its true value, every update step strictly contracts errors by factor ρ.

Practical blueprint for metrologists. To verify Axiom 3.1 in a given instrument:

1. *Choose your statistic T* (e.g. mean and sd, regression vector, kernel estimator).
2. *Empirically estimate* or theoretically compute the Jacobian $\partial_\theta U(\theta, z)$ at θ_Q (using calibration data or model linearization).
3. *Find a neighborhood $B_r(\theta_Q)$* on which the spectral radius $\|\partial_\theta U\| < \rho < 1$.
4. *Validate Lipschitz constants L_E, L_R* for emission and region-map via small perturbation tests.

Once these steps confirm $\rho < 1$, your system satisfies the True-Value Axiom and hence admits a finite-horizon guard-band certification.

Now we have exactly the regularity conditions needed to state and prove the Guard-Band Equivalence in what follows.

Theorem 3.1 (The Guard-Band Equivalence)**.** *A tail predicate P_{tail} admits a universal, polynomial-size deterministic witness for every possible head of the stream if and only if it is logically equivalent to a guard-band predicate.*

3.3 Finite-Horizon Truncation Lemma

Lemma 3.1 (Finite-Horizon Truncation (Local Contraction))**.** *Under the assumptions of the previous subsection, let*

$$\delta \;=\; \mathit{margin}\big(\mathcal{R}_{\theta_Q}\big) \;>\; 0,$$

and choose the horizon

$$T \;=\; \min\Big\{\, t : \rho^t\, r\, L_E\, L_R < \delta \Big\}.$$

Then for any initial statistic $\theta_n \in B_r(\theta_Q)$, verifying

$$E\big(\theta_{n+k}\big) \in \mathcal{R}_{\theta_Q} \quad \text{for } k = 1, 2, \ldots, T$$

implies

$$E\big(\theta_{n+k}\big) \in \mathcal{R}_{\theta_Q} \quad \text{for all } k \geq 1.$$

That is, checking only the next T steps suffices to guarantee membership forever.

Proof. Since $\theta_n, \hat{\theta} \in B_r(\theta_Q)$ and U is ρ-contractive there,

$$d_\Theta\big(\theta_{n+k}, \hat{\theta}_{n+k}\big) \;\leq\; \rho^k\, d_\Theta(\theta_n, \hat{\theta}) \;\leq\; \rho^k\, r.$$

By emission-Lipschitz and region-map Lipschitz, a deviation of $\rho^k r$ in Θ translates to at most $L_E\, L_R\, \rho^k r$ in the Hausdorff sense on $\mathcal{R}$. Choosing T so that $L_E\, L_R\, \rho^T r < \delta$ ensures no further outputs can escape $\mathcal{R}_{\theta_Q}$ once the first T are inside. □

3.4 Efficient Covering Lemma

Lemma 3.2 (Efficient Covering of Θ)**.** *Let $\Theta \subset \mathbb{R}^d$ be compact with diameter D. For any precision $\varepsilon > 0$, one can in time poly$\big(d, \log(D/\varepsilon)\big)$ construct an explicit ε-net $\{\eta_1, \ldots, \eta_N\} \subset \Theta$ of size $N = O\big((D/\varepsilon)^d\big)$, such that $\bigcup_{j=1}^N B(\eta_j; \varepsilon) = \Theta$.*

Proof. **1. Grid in ambient space.** Partition each coordinate into intervals of length $\varepsilon/\sqrt{d}$, yielding $M = O((D/\varepsilon)^d)$ grid points.

2. Projection or pruning. For each grid point, test membership in Θ (via the known description of Θ) in poly$(d, \log(1/\varepsilon))$ time. Keep those that lie in Θ.

3. Complexity. There are $O((D/\varepsilon)^d)$ points, each checked in poly$(d, \log(D/\varepsilon))$ time, hence total time is poly$(d, \log(D/\varepsilon))$. □

3.5 (⇐) Sufficiency: Guard-Bands are Certifiable

This direction demonstrates that if a property has the guard-band structure, it is indeed testable. The proof formalizes the constructive argument from Chapter 5.

Lemma 3.3 (Region-Approximation)**.** *Under assumption (A5), the map $\theta \mapsto \mathcal{R}_\theta$ satisfies*

$$d_H\big(\mathcal{R}_\theta,\, \mathcal{R}_{\theta'}\big) \;\le\; L_R\, d_\Theta(\theta, \theta'),$$

where d_H is the Hausdorff metric on subsets of $\mathcal{Z}$. In particular, if $d_\Theta(\theta, \theta_Q) \le \varepsilon$, then

$$\mathcal{R}_\theta \;\subseteq\; \mathcal{R}_{\theta_Q}^{+L_R\varepsilon} \quad \textit{and} \quad \mathcal{R}_{\theta_Q} \;\subseteq\; \mathcal{R}_\theta^{+L_R\varepsilon}.$$

Proof of Sufficiency. Assume P_{tail} is a guard-band predicate with statistic map T and regions $\{\mathcal{R}_\theta\}$. Let $\theta_Q = T(P_Q)$.

(i) Choose precision ε. Let

$$\delta \;=\; \operatorname{margin}\big(\mathcal{R}_{\theta_Q}\big) > 0,$$

and fix

$$\varepsilon \;=\; \min\Big(r,\; \frac{\delta}{2\, L_E\, L_R}\Big),$$

where $r, \rho < 1$ come from (A6) and L_E, L_R from (A3),(A5).

(ii) Build the net and pick the net-point. By Lemma 3.4, in time $\operatorname{poly}(d, \log(1/\varepsilon))$ we construct an ε-net $\{\eta_1, \ldots, \eta_N\} \subset \Theta$. Since $\theta_Q \in \Theta$, there is some η_j with $\|\eta_j - \theta_Q\| \le \varepsilon$. We will use j as our witness.

(iii) Compute the finite horizon. By Lemma 3.3 (with initial error $\le \varepsilon$), choose the minimal

$$T \;=\; \min\Big\{\, t : \rho^t\, \varepsilon\, L_E\, L_R < \delta \Big\} \;=\; O\big(\log(1/\varepsilon)\big).$$

(iv) The witness. The prover's certificate is the single integer

$$w = j,$$

encoded in $\log_2 N = O(\log(1/\varepsilon))$ bits. By construction $\eta_j \in B_r(\theta_Q)$.

(v) Verification. On input $(z_{1:n}, w = j)$:

1. Decode j to get $\widehat{\theta} = \eta_j$.
2. *Simulate* the update map U starting from $\widehat{\theta}$ for exactly T steps, producing $\widehat{\theta}_{n+1}, \ldots, \widehat{\theta}_{n+T}$.
3. For each $k = 1, \ldots, T$, compute

   $$z' = E\big(\widehat{\theta}_{n+k-1}\big) \quad \text{and check} \quad z' \;\in\; \mathcal{R}_{\widehat{\theta}}.$$

   Then invoke Lemma 3.3: since $d_\Theta(\widehat{\theta}, \theta_Q) \le \varepsilon$ implies $\mathcal{R}_{\widehat{\theta}} \subseteq \mathcal{R}_{\theta_Q}$, we conclude

   $$z' \in \mathcal{R}_{\theta_Q}.$$

   If any of these refined checks fails, *reject*.

4. Otherwise *accept*.

Each simulation-and-check step takes poly(n) time by (A2),(A3),(A5), and there are only $T = O(\log(1/\varepsilon))$ of them.

(vi) Correctness by Finite-Horizon. Since $\|\widehat{\theta} - \theta_Q\| \leq \varepsilon \leq r$, Lemma 3.3 guarantees that if all of the first T simulated outputs lie in $\mathcal{R}_{\theta_Q}$, then *all* future outputs will. Thus acceptance exactly characterizes $\forall k > n : \; Z_k \in \mathcal{R}_{\theta_Q}$, i.e. P_{tail}.

Because $|w| = O(\log(1/\varepsilon))$ and everything runs in poly(n) time, this puts the language of good heads in **NP**. □

3.6 (⇒) Necessity: Certifiability Implies a Guard-Band

This is the more profound direction, for which we now provide a full, rigorous proof. We assume a certification system exists and deduce its necessary structure.

Proof of Necessity. We work in the measurement-chain model of Chapter 4: $Q \xrightarrow{f} X \xrightarrow{g} Y \xrightarrow{\text{Generate}} Z_{1:\infty}$, where Q is the unknown true state and $Y = g(f(Q))$ is the finite instrument state. An admissible predicate P_{tail} is one decided by a finite, polynomial-time test Φ on a block of the tail, where the test is robust to ε-perturbations of the initial state Q.

The NP-Certifiability Assumption. We assume P_{tail} is in **NP**. This means there exists a polynomial $p(n)$ and a deterministic, polynomial-time verifier $V(z_{1:n}, w)$ such that for every head $z_{1:n}$ arising from a true state Q, the predicate holds if and only if there exists a witness w with $|w| \leq p(n)$ that causes V to accept.

Because V runs in time polynomial in n, it cannot inspect an infinite tail. Its logic must rely on the admissibility of the predicate. Any correct witness w must allow V to:

1. Reconstruct a candidate instrument state Y^* (which corresponds to an ε-approximation of the true state Q^*) from the witness w.
2. Simulate the generator for a finite, polynomial number of steps $T(n)$, producing a predicted tail block $\widehat{z}_{n+1:n+T(n)}$.
3. Evaluate the Boolean test $\Phi(\widehat{z}_{n+1:n+T(n)})$ and accept if and only if it is true.

No other verification strategy can succeed in polynomial time.

Lemma 3.4 (Block-to-Symbol Guard-Band Reduction)**.** *Let Φ be a deterministic predicate on length-T blocks over alphabet $\mathcal{Z}$, and suppose the verifier accepts an infinite tail $z_{n+1}z_{n+2}\dots$ if and only if for every $k \geq n+1$, the length-T block*

$$B_k := z_k z_{k+1} \dots z_{k+T-1}$$

satisfies $\Phi(B_k) = \text{True}$. Define the per-symbol region

$$\mathcal{R} := \left\{ z \in \mathcal{Z} \;\middle|\; \forall u \in \mathcal{Z}^{T-1},\; \Phi(zu) = \text{True} \right\},$$

where zu denotes the concatenation of symbol z with any length-$(T-1)$ continuation u. Then any tail accepted by the verifier must satisfy

$$\forall k > n, \quad z_k \in \mathcal{R}.$$

In other words, acceptance implies the per-symbol guard-band condition $z_k \in \mathcal{R}$ for all future positions.

Proof. Suppose, for contradiction, that an infinite tail $z_{n+1}z_{n+2}\dots$ is accepted by the verifier but there exists some $k > n$ with $z_k \notin \mathcal{R}$. By definition of $\mathcal{R}$, since $z_k \notin \mathcal{R}$, there exists a continuation $u \in \mathcal{Z}^{T-1}$ such that $\Phi(z_k u) = \text{False}$. Let $B_k = z_k z_{k+1} \dots z_{k+T-1}$ be the actual length-T block appearing in the tail at position k.

The verifier's acceptance criterion requires $\Phi(B_k) = \text{True}$, because it accepts only if *every* sliding block of length T passes Φ. But since Φ is deterministic, and u is some length-$(T-1)$ string for which $\Phi(z_k u) = \text{False}$, the only way to avoid rejection is the actual continuation $z_{k+1} \dots z_{k+T-1}$ differs from u. However, the tail is arbitrary beyond the assumed acceptance, and the fact that there exists even one such bad extension u means that the verifier would have to rule out an infinite family of tails in advance to guarantee correctness. Since acceptance must hold for the given tail, and no further mechanism is provided to "repair" a symbol that has an unsafe possible extension, the only consistent possibility is that z_k has no such bad extension; that is, $z_k \in \mathcal{R}$. This contradicts the assumption $z_k \notin \mathcal{R}$.

Thus every symbol in an accepted tail must lie in $\mathcal{R}$, establishing the per-symbol guard-band condition. □

Step 1: The Witness Defines a Guard-Band. We now convert this NP-scheme into a guard-band predicate. For each possible witness string w of length $\leq p(n)$, we can define an acceptance region $\mathcal{R}_w \subseteq \mathcal{Z}$. Since the verifier V is deterministic and streaming, and must halt on the first violation, its logic for a given witness w must be equivalent to a per-sample membership test. While the admissibility test Φ may look at a block of $T(n)$ symbols, for the overall infinite-tail property to hold, every such block (starting at $n+1, n+2, \dots$) must be accepted (Lemma 3.4). This forces the property to be a per-sample one.

We can define $\mathcal{R}_w$ as the set of all $z \in \mathcal{Z}$ such that if z were to appear in any position in the tail, it would not cause the verifier to reject. Formally:

$$\mathcal{R}_w = \{z \in \mathcal{Z} \mid \text{For all tail prefixes } u, \Phi(u \cdot z \cdot \dots) \text{ remains true}\}$$

By construction, $\mathcal{R}_w$ is a fixed subset of $\mathcal{Z}$ whose description complexity is polynomial.

Step 2: The Equivalence to a Guard-Band Predicate. Let θ be the information encoded in the witness w. For a true head $z_{1:n}$ arising from a state Q:

1. If there is an accepting witness w, then V's internal simulation must have passed, meaning $\Phi(\hat{z}^{(w)}_{n+1:n+T(n)}) = \text{true}$. By the ε-robustness of admissible predicates, the actual tail $(z_{n+1}, \dots)$ must also satisfy the property. By the construction of $\mathcal{R}_w$, this implies that $\forall k > n, z_k \in \mathcal{R}_w$. So we have $\forall k > n, z_k \in \mathcal{R}_\theta$.

2. Conversely, if the true tail stream satisfies $\forall k > n, z_k \in \mathcal{R}_\theta$ for the θ corresponding to the witness w for the true state Q, then the verifier's simulation will pass, and the witness w will be accepted.

This establishes the equivalence: $z_{1:n} \in L_P \iff \exists w$ s.t. $\forall k > n, z_k \in \mathcal{R}_{\theta_w}$. The statistic $T(P_Q)$ is simply the map that takes the underlying distribution to the correct "coarse-label" θ that the prover must witness.

Step 3: The Lipschitz Continuity. Finally, we must argue that the maps $T : P_Q \mapsto \theta$ and $\theta \mapsto \mathcal{R}_\theta$ are Lipschitz.

- The map from the true state Q to the instrument state Y (our coarse label θ) is $g \circ f$. Our Model Regularity condition already assumes this map is continuous (and in fact, finite-to-one, a stronger condition).

- The map from the coarse label θ to the acceptance region $\mathcal{R}_\theta$ is also Lipschitz. The region $\mathcal{R}_\theta$ is defined by the behavior of the deterministic simulator and the Boolean test Φ. Since the simulator and Φ are assumed to be polynomial-time computable (and thus continuous in their inputs), a small change in the initial state θ will result in a small, uniformly bounded change in the predicted tail block, which in turn results in a small change to the boundary of the set $\mathcal{R}_\theta$. This small, uniform change is precisely the definition of Lipschitz continuity in the Hausdorff metric for sets.

This completes the necessity proof. We have shown that any NP-certifiable, admissible tail predicate must be equivalent to a guard-band predicate with the required Lipschitz properties. □

4 Conclusion

This completes our characterization of deterministic testability. The Guard-Band Equivalence Theorem provides a definitive link between epistemology (what is certifiable) and physics (the structure of the measurement system). It establishes that for a statement about a measurement's infinite future to be knowable with certainty from a finite proof, the system must lose information in a structured way. This loss creates a finite-dimensional state, which in turn defines a set of possible "lanes" or "guard-bands" for the system's future behavior. The witness simply serves to tell the verifier which lane the system is in.

In Chapter 7, we will unify this deterministic view with the probabilistic confidence paradigm, showing that Guard-Band predicates arise as the zero-error limit of statistical confidence intervals.

Chapter 7: From Statistical Confidence to Logical Certainty: A Unification of Frameworks

Dmytro Surdu

Our theory has so far focused on the conditions for absolute, deterministic proof, culminating in the Guard-Band Equivalence theorem. This world of logical certainty seems distinct from the everyday practice of metrology, which is grounded in statistical inference and statements of confidence. This chapter bridges that gap. We will demonstrate that these two perspectives are not for different systems, but are different ways of knowing the ***same system****. The demand for absolute certainty forces a transition from a black-box statistical view to a mechanistic one, revealing that the Guard-Band structure is the necessary internal logic of any certifiable measurement.*

1 One Object, Two Epistemological Intents

At its heart, any repeatable measurement can be seen as generating a **stationary stochastic process**—a stream of data, $\{Z_k\}_{k=1}^{\infty}$, whose statistical properties do not change over time. The simplest such process is a sequence of independent and identically distributed (i.i.d.) random variables, which forms the basis of elementary statistics. Our deterministic chain model, $Z = h(Q)$, when in its stable attractor state, also generates a stationary process.

The crucial insight, and the central thesis of this chapter, is that the difference between standard statistical analysis and our complexity-theoretic framework is not in the nature of the underlying physical process, but in the **epistemological intent** of the observer.

- **Intent 1: Statistical Confidence ($\delta > 0$).** The observer's goal is to make a statement that is *probably* true. A small, non-zero risk of error, δ, is acceptable. This intent permits a "black-box" view of the process. We do not need to know the internal mechanism; we only need to draw a sample and analyze it, relying on the law of large numbers. The evidence is the data itself.

- **Intent 2: Logical Certainty ($\delta = 0$).** The observer's goal is to make a statement that is *provably*, absolutely true, with zero risk of error. This intent is far more demanding and, as we will see, forces a change in perspective.

This chapter will show that the shift in intent from $\delta > 0$ to $\delta = 0$ is not merely quantitative but qualitative, compelling a transition from a statistical view to a structural, mechanistic one.

2 The Black-Box View: Verification with Confidence ($\delta > 0$)

When we are content with confidence rather than certainty, we treat the measurement system as a black box generating i.i.d. samples $X_1, \ldots, X_n$ from an unknown distribution P_X. Our goal is to make a decision about a property of the form $\Phi(P_X) \in A$, where Φ is a functional of the distribution (e.g., the mean or variance).

Since we cannot know P_X, we estimate it with the empirical distribution $\widehat{P}_n = \frac{1}{n}\sum_{i=1}^{n} \delta_{X_i}$. The core question becomes: for which functionals Φ is the estimate $\Phi(\widehat{P}_n)$ a reliable proxy for the true value $\Phi(P_X)$?

Definition 2.1 (Polynomial-Decidable Functional)**.** A functional Φ is **polynomial-decidable** if, for any desired precision $\varepsilon > 0$ and confidence level $1-\delta$, the number of samples n required to ensure

$$\Pr\left(|\Phi(\widehat{P}_n) - \Phi(P_X)| \leq \varepsilon\right) \geq 1 - \delta$$

is a polynomial function of $1/\varepsilon$ and $\log(1/\delta)$.

The key property that enables such efficient estimation is typically a form of continuity.

Interpretive Remark. What follows is a heuristic, high-level interpretation—not a formal theorem. It illustrates how vanishing statistical uncertainty can be viewed as suggestive of an underlying guard-band structure, but no formal derivation later depends on this intuition. The rigorous statements and equivalences remain those proved elsewhere.

Theorem 2.1 (Continuity enables Statistical Estimation (Informal))**.** *If a functional Φ is Lipschitz continuous with respect to a metric d on probability distributions, i.e., $|\Phi(P) - \Phi(Q)| \leq L \cdot d(P, Q)$, and the empirical convergence rate is known (e.g., $d(\widehat{P}_n, P_X) \approx O(n^{-1/2})$), then Φ is polynomial-decidable.*

Sketch. The Lipschitz condition ensures that the error in the estimate is bounded by the error in the empirical distribution: $|\Phi(\widehat{P}_n) - \Phi(P_X)| \leq L \cdot d(\widehat{P}_n, P_X)$. Since the statistical error $d(\widehat{P}_n, P_X)$ can be made smaller than any ε/L with a polynomially large sample size n, the error in the functional estimate can also be bounded by ε. □

In this regime, verification is probabilistic. We tolerate the risk δ that our finite sample was "unlucky" or unrepresentative.

3 The Mechanistic View: Verification with Certainty ($\delta = 0$)

Now, consider the profound consequences of demanding absolute certainty by setting $\delta = 0$.

The black-box perspective immediately becomes untenable. The possibility of an "unlucky" sample is no longer a small risk to be managed; it is a fatal counterexample that must be ruled out entirely. A finite sample from a black box can never provide a definitive proof

about the properties of the infinite stream, because the very next sample is an unconstrained random event that could violate the claim.

To achieve certainty, the observer is **forced to abandon the black-box view**. We must "open the box" and analyze the underlying mechanism that generates the data. We must be able to prove that a violating sample is not just improbable, but *impossible* given the system's determined state.

This forces a fundamental change in the model of the process. It cannot be just any stationary process; it must be one whose randomness is structured in a very specific way.

The next observation is that, under the guard-band certification model (which specifies the class of admissible tail predicates, finite-length witness format, and polynomial-time verification), the following holds:

Theorem 3.1 (The Structural Requirement for Certainty)**.** *A stationary stochastic process allows for a property to be certified with logical certainty ($\delta = 0$) from a finite witness if and only if the process is generated by a deterministic mechanism $Z = h(Q)$, where all randomness is front-loaded into the initial state Q, and the structure of the function h is equivalent to a Guard-Band model.*

4 The Guard-Band as the Limit of a Confidence Interval

The connection between the two views becomes clear when we trace what happens to a statistical confidence interval as we push the demand for confidence to its logical extreme.

1. **The Statistical Confidence Interval ($\delta > 0$):** We have a confidence interval $\mathcal{C}_n$ for the property $\Phi(P_X)$. It is centered on our data-derived estimate $\widehat{\Phi}_n$ and has a width β_n that reflects our uncertainty. It is a statement about the *property*, based on the *data*.

2. **Forcing Certainty ($\delta \to 0$):** To achieve certainty, we need the width of our confidence interval to be zero ($\beta_n \to 0$) and the probability of error to be zero. A finite number of samples from a black box cannot satisfy this. The only way to meet this demand is if the system's internal structure itself provides the guarantee.

3. **The Emergence of the Guard-Band:** The demand for certainty forces a conceptual transformation:

   - The property being measured (described by the functional Φ) and the evidence (the data sample) must merge into a structural property of the system itself (described by the statistic T and the guard-map $\mathcal{R}$).
   - The probabilistic confidence interval $\mathcal{C}_n$, which is a property of our *knowledge*, is replaced by the deterministic acceptance region $\mathcal{R}_{\theta_Q}$, which is a property of the *system's state*.

Remark 4.1 (The Limit of a Confidence Interval)**.** The Guard-Band is the structural reification of a confidence interval that has been forced to have zero width and zero error probability. It is the object that remains when a guarantee becomes so strong that it can no longer be based on the randomness of data, and must instead be based on the deterministic laws governing the data generator.

5 Conclusion: A Unified View of Measurement Knowledge

The statistical and complexity-theoretic frameworks are not for different kinds of systems; they are for different kinds of questions about the same systems. The table below summarizes this unified view.

Table 1: Two Intents for Verifying a Property of a Stationary Process

Aspect	**Statistical Confidence** ($\delta > 0$)	**Logical Certainty** ($\delta = 0$)
View of System	External, "Black-Box"	Internal, Mechanistic
Model of Randomness	Ongoing, i.i.d. draws	Front-loaded, Structural (Q)
Source of Trust	Law of Large Numbers	Provable System Structure
Key Enabler	Lipschitz continuity of Φ	Guard-Band structure of $(h, T, \mathcal{R})$
Evidence	The data sample ($X_{1:n}$)	The state witness ($w = (y, r)$)
Conclusion	Probabilistic (Confidence)	Deterministic (Proof)

Our theory proves that the mechanistic view required for certainty must necessarily conform to the Guard-Band model. This provides a deep and satisfying unity to the science of measurement. It shows that the logical certainty sought by complexity theory is the rigorous and necessary endpoint of the journey from the probabilistic confidence sought by statistics. It fills the conceptual gap between possessing statistical information and stating a logical fact.

Chapter 8: Conclusion: A New Foundation and Its Open Questions

Dmytro Surdu

We have traveled from the initial observation that verifying a claim is often easier than finding its proof, through the formalisms of complexity and information theory, to a complete structural characterization of certifiable measurement. This concluding chapter will summarize the central thesis of this work, delineate precisely what has been solved, and outline the new horizons of research that this foundation makes possible.

1 The Theory in a Nutshell

This textbook set out to answer a fundamental question: under what conditions can a finite, deterministic proof be given for a property of an infinite measurement stream? After exploring the limits of certification and developing the necessary mathematical tools, we arrived at a powerful and comprehensive answer, summarized by the following chain of logical equivalence:

Deterministic Testability
($\Updownarrow$)
A property admits a universal, polynomial-size, deterministic witness.
($\Updownarrow$) *(The Guard-Band Equivalence, Ch. 6)*
Guard-Band Structure
($\Updownarrow$)
The property is equivalent to $\forall k, Z_k \in \mathcal{R}_{T(P_Q)}$.
($\Updownarrow$) *(Structural Implication)*
Finite State and Information Loss
($\Updownarrow$)
The underlying measurement model $h(Q)$ is equivalent to a system with a finite-dimensional internal state created by an information-losing transformation.
($\Updownarrow$) *(Physical Prerequisite)*
Model Regularity (UFC)
The model's uncertainty structure is non-pathological and uniformly bounded.

This chain represents the core contribution of our work. It forges an unbreakable link between an epistemological requirement (testability) and a set of physical and structural

properties of the measurement system itself. We have shown not only that information loss and finite state are sufficient to enable certification, but that they are, in a deep and formal sense, necessary for it.

2 Implications for Theoretical Metrology

This framework provides a new, rigorous language for discussing concepts that have long been central to measurement science.

- **A Formal Definition of "Measurand":** The traditional concept of a measurand can be formalized through the lens of our theory. A property is a "well-defined measurand" in a deterministic sense only if it corresponds to a guard-band predicate. This provides a sharp criterion to distinguish between properties that are fundamentally certifiable and those that can only be described statistically.

- **A New Look at Model Validity:** The framework provides a new approach to model validation. Instead of just checking for statistical consistency, one can analyze the model's structural properties. Does the proposed model for a measurement system possess the necessary information-losing structure and satisfy the Model Regularity Condition (UFC)? If not, our theory predicts that no amount of testing can provide absolute certainty about its long-term behavior.

- **Understanding Type A Uncertainty:** Our results provide a solid theoretical foundation for Type A uncertainty evaluation. The process of taking samples to estimate a mean can be seen as a statistical procedure. Our theory shows that the logical certainty one desires in the final stated result is only possible because the underlying "ideal" process converges to the guard-band structure as the demand for confidence becomes absolute ($\delta = 0$).

3 What Has Been Solved

This work provides a complete, self-contained theory for a specific but fundamental domain. We can definitively state that the following has been solved:

1. **The Conditions for Deterministic Certification:** We have provided the necessary and sufficient conditions under which a property of a deterministic measurement stream can be certified by a short, deterministic witness.

2. **The Class of Certifiable Models:** We have characterized the exact class of measurement models that allow for such certification. They are equivalent to finite-state transducers that lose information in a structured, non-pathological way.

3. **The Foundation for Type A Analysis:** We have provided a logical underpinning for the transition from statistical sampling (Type A evaluation) to a statement of fact about a measurand, showing that the latter requires the system to possess a specific deterministic structure.

4 Open Questions and Future Horizons

A new foundation invariably reveals new territories to explore. Our theory, while complete in its domain, opens several exciting avenues for future research.

4.1 Open Problem 1: A Theory for Type B Uncertainty

Our framework is built on an initial state Q drawn from a distribution P_Q, which is then processed. This is analogous to Type A uncertainty, which arises from statistical variation. The great challenge is to build a corresponding theory for Type B uncertainty—information derived from other sources, such as calibration certificates, physical laws, or expert judgment.

- **Hypothesis:** A theory for Type B might require moving from a single random variable Q to a *distribution over distributions*, or a Bayesian framework where the state is a probability distribution itself. The transition function 'h' might need to become probabilistic, mapping distributions to distributions.
- **The Question:** What are the necessary and sufficient structural properties of such a probabilistic model that would allow for certification?

4.2 Open Problem 2: From Theoretical to Constructive Foundations

This textbook has laid out a theoretical foundation. The next step is to use this theory to inform the design of real-world systems. This involves translating the abstract concepts of Guard-Bands, states, and models into a concrete, practical software architecture.

- **The Goal:** To develop a "Measurement Test Architecture" (MTA), as hinted at in the original engineering work that motivated this theory. Such an architecture would provide a set of programming constructs and design patterns that *guarantee by construction* that any measurement procedure built with them is certifiable.
- **The Challenge:** How do you design programming language features that directly correspond to guard-bands, state transitions, and information-losing models, and that can be statically analyzed and formally verified?

4.3 Open Problem 3: Beyond Determinism: Interactive Proofs

Our work has focused on the limits of *deterministic* witnesses. But what about properties that fall outside the Guard-Band class? Are they simply unknowable? Complexity theory offers a richer set of verification models.

- **The Idea:** An **interactive proof system** (leading to complexity classes like **AM** or **IP**) allows a probabilistic verifier to interact with an all-powerful prover by asking a series of random questions. This interaction can often certify properties that have no short deterministic witness.

- **The Question for Metrology:** Could we design an interactive protocol between a metrologist (the verifier) and a complex simulation or physical system (the prover) to validate statistical claims like "this data is Gaussian" or "this process is ergodic"? What would such a protocol look like?

These open questions show that the application of complexity-theoretic ideas to the foundations of measurement is a rich and fertile field of inquiry. The work presented in this textbook provides the first rigorous step, a solid foundation upon which new and more comprehensive theories of measurement knowledge can be built.

Appendix: Detailed Proof of Π^0_2-Hardness for Lossless Channels

Dmytro Surdu

Necessity of Information Loss via Reduction from TOTAL

Definition (The TOTAL Problem)**.** Let $\{M_e\}_{e\in\mathbb{N}}$ be a standard enumeration of Turing machines. The set

$$\text{TOTAL} \;=\; \big\{\, e \in \mathbb{N} : \; \forall x \in \mathbb{N}\; \exists t \; M_e(x) \text{ halts within } t \text{ steps} \big\}$$

is Π^0_2–complete.

We now show that for any lossless (invertible) discrete measurement channel, the corresponding infinite–tail certification problem is at least as hard as TOTAL. In particular, no finite deterministic witness can collapse this to an **NP** decision problem.

Definition (Discrete Lossless Measurement Model)**.** We work in the following extremely simple, purely *discrete* model:

- **State space:** $\mathcal{Q} = \mathbb{N}$.
- **Channel:** $g \circ f : \mathcal{Q} \to \mathcal{Q}$ is the identity map, which is a bijection (hence lossless).
- **Stream generator** h**:** For each initial state $q = e \in \mathbb{N}$, $h(e)$ outputs an infinite sequence over the alphabet $\{1, \#, \perp\}$ by dovetailing simulations of $M_e(1), M_e(2), \ldots$ as follows:
  1. Interleave one step of $M_e(1)$, then one step of $M_e(2)$, then one step of $M_e(1)$, then one step of $M_e(3)$, and so on, ensuring every $M_e(x)$ makes progress.
  2. Whenever $M_e(x)$ is discovered to halt in t steps, immediately append
  $$1^x \,\#\, 1^t \,\#$$
  to the output stream, then resume the dovetailed simulation.
  3. If $M_e(x)$ never halts, no certificate for that x ever appears.
  4. After every halting event the process continues indefinitely, so the overall output is infinite.

Definition (Tail Predicate P_e)**.** For each $e \in \mathbb{N}$, define the infinite–tail predicate P_e on streams $Z \in \{1, \#, \perp\}^\omega$ by

$$P_e(Z) \quad : \quad \forall x \in \mathbb{N}\ \exists t \in \mathbb{N} \text{ s.t. } 1^x\#1^t\# \text{ appears as a contiguous substring of } Z.$$

Theorem. *Under the discrete lossless model above, deciding whether a given stream satisfies P_e is Π^0_2–hard. Hence, without information loss (invertibility), the certification problem lies outside* **NP** *unless the Polynomial Hierarchy collapses.*

Proof. We give a many–one reduction from TOTAL to the certification problem for P_e.

(i) Correctness of the reduction. Let $e \in \mathbb{N}$ be arbitrary. Consider the stream

$$Z \;=\; h(e) \;=\; h_e(e).$$

By construction:

- $\Rightarrow$ If $e \in$ TOTAL, then for every $x \in \mathbb{N}$, the machine $M_e(x)$ halts in some finite number of steps t_x. During the dovetailed simulation of all $M_e(x)$, each halting event is eventually discovered, and so for each x the certificate block $1^x\#1^{t_x}\#$ is appended to Z. Consequently $\forall x\, \exists t\, \big(1^x\#1^t\# \text{ appears in } Z\big)$, i.e. $P_e(Z)$ holds.
- $\Leftarrow$ Conversely, if $P_e(h(e))$ holds, then for every x there is some t such that $1^x\#1^t\#$ appears in the output. By the definition of h, the only way such a block can ever be produced is upon discovering that $M_e(x)$ indeed halts in exactly t steps. Hence $M_e(x)$ halts for all x, and so $e \in$ TOTAL.

Thus

$$e \in \text{TOTAL} \iff P_e\big(h(e)\big).$$

(ii) Complexity implications. Because $g \circ f$ is the identity on $\mathbb{N}$, a finite observation of the stream's "head" gives *exactly* the initial state e. There is no loss of information at the channel stage. Any deterministic verifier that, given a finite witness and a finite prefix of the stream, tries to decide P_e must effectively solve

$$\forall x\ \exists t\ M_e(x) \text{ halts in } t \text{ steps},$$

which is a canonical Π^0_2–complete problem. No finite, polynomial–size witness can encode "all future halts" into a single certificate checkable in polynomial time. Hence the certification language cannot lie in **NP** unless the Polynomial Hierarchy collapses.

$\square$